

1. **What is a data type in context in the context of programming and operating systems?**

A data type defines the format, size, and interpretation of values a variable can hold in memory.

2. **How many bytes does the OS allocate for a char data type? How many bytes for a double?**

A char is allocated 1 byte, and a double is allocated 8 bytes.

3. **Name the five main memory segments in a process memory layout.**

The five main segments are the Text segment, Data segment, BSS segment, Stack, and Heap.

4. **What is the difference between the Data segment and the BSS segment?**

The Data segment stores initialized global variables, while the BSS segment stores uninitialized global variables.

5. **Why does the OS place the stack at high virtual addresses and the text segment at low addresses?**

The stack is placed at high addresses to grow downward, while the text segment is at low addresses.

6. **What happens to stack memory when a function returns? Who reclaims it?**

Stack memory is automatically reclaimed by the CPU/runtime environment when a function returns.

7. **What is the risk of forgetting to call free() on heap memory?**

Forgetting to call free() on heap memory results in a memory leak.

8. **Explain the difference between ptr and &ptr when printing addresses.**

ptr gives the address stored inside the pointer, whereas &ptr gives the address of the pointer variable itself.

9. **Why is the cast (void *) used when printing pointers with %p?**

(void *) is used with %p to ensure type compatibility for printing generic pointers.

10. **How does the OS know which memory belongs to which process?**

The OS tracks process memory ownership using the Memory Management Unit (MMU) and page tables.